Friedrich-Alexander-Universität
Technische Fakultät

Pattern Recognition Lab

FAU

Project Time Series (SS 26)

Detecting Parkinson's Disease from Gait Signals with Mamba

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Gait carries a measurable signature of PD

Introduction & motivation

Clinical Need

- 2nd Most common neurodegenerative disease; gait is one of its earliest motor signs
- Clinical scales (UPDRS) depend on expert rating and are coarse
- Sensor data:
- Instrumented insoles: 8 force sensors per foot, 100 Hz

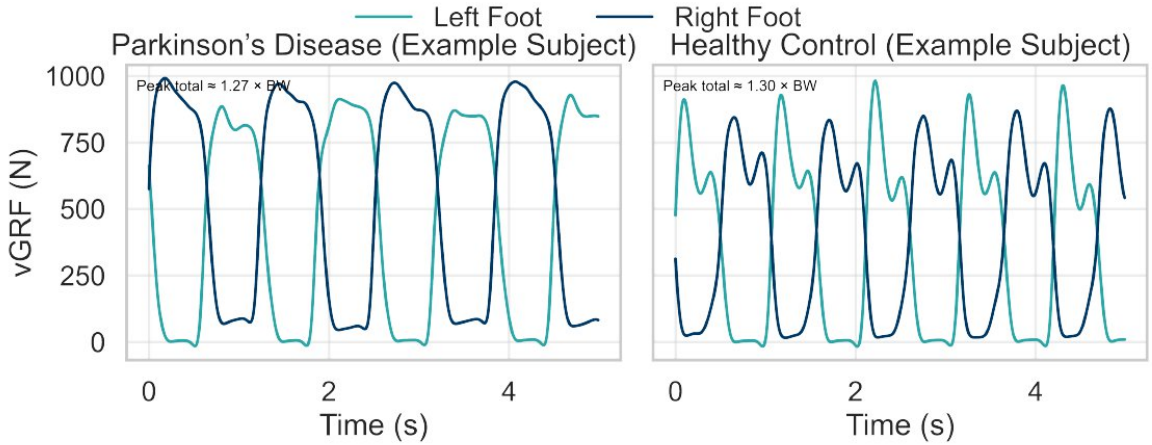


Fig. 1 *Five seconds of vGRF: a PD patient and a healthy control*

From hand-crafted features to SSMs

State of the art

Classical ML	Deep learning	State-space models
• Hand-crafted statistics of VGRF channels (e.g. TSFresh features) • Random Forest, SVM • Strong on small cohorts; interpretable	• 1D-ResNet [1], RNN / LSTM, transformers • Raw signal or feature sequences • Reported accuracies up to 96–99 % [6, 7]	• Structured State Space Sequence model (S4) [2] → Mamba [3]: input-dependent (selective) state transitions • Linear time in sequence length • Used for speech, EEG, ECG

- Reported accuracies span 70–99 %, which should we believe?

[1] He et al., CVPR 2016 [2] Gu et al., ICLR 2022 [3] Gu & Dao, arXiv:2312.00752, 2023 [6] Diagnostics 15(16):2046, 2025 [7] Sensors, 2021 (PMC8347971)

Can Mamba improve PD detection from gait?

Research question & objectives

Research question (project aim)

Can Mamba models improve the performance of classification models trained to detect Parkinson's disease using gait signals?

1 – Mamba Models

- Build a Mamba (selective state-space) classifier on the VGRF signals of 100 subjects

2 – Baseline Models

- Compare it with Random Forest and SVM (RBF), plus 1D-ResNet and an LSTM on the same input, on identical subject-wise 5-fold CV

3 – Architecture

- Isolate what drives performance: input representation, encoder capacity, scan direction

Why Mamba Should Fit Gait

Research question & objectives

Hypothesis: the PD signature is temporal – stride-time variability, irregular timing, left–right asymmetry across many steps [9]



1 s feature (198 statistics) → 60 per window

■ irregular seconds (illustration)

Random Forest / SVM	LSTM (same features)	Mamba (same features)
• One feature vector per window • Cannot relate an irregular step to its context	• Reads the ordered sequence • Fixed gated recurrence, sequential training	• Selection: weight irregular seconds, pass over steady walking

[2] Hausdorff, Chaos 19(2):026113, 2009

100 subjects, matched on everything except gait

Materials & methods – data

	HC (n = 50)	PD (n = 50)	p
Female / male	25 / 25	25 / 25	1.00
Age (years)	63.0 ± 8.4	66.5 ± 9.8	0.06
Height (cm)	167.6 ± 8.6	166.3 ± 7.9	0.45
Weight (kg)	71.9 ± 12.8	68.5 ± 12.0	0.18
BMI (kg/m²)	25.5 ± 3.7	24.7 ± 3.1	0.26
Gait speed (m/s)	**1.24 ± 0.17**	**1.02 ± 0.22**	**< 0.001**
TUAG (s)	**9.2 ± 1.6**	**12.2 ± 5.0**	**< 0.001**
Hoehn & Yahr	–	2.3 ± 0.4	–
UPDRS (total)	–	29.5 ± 11.1	–
Study Ga / Ju / Si	13 / 16 / 21	18 / 16 / 16	0.48

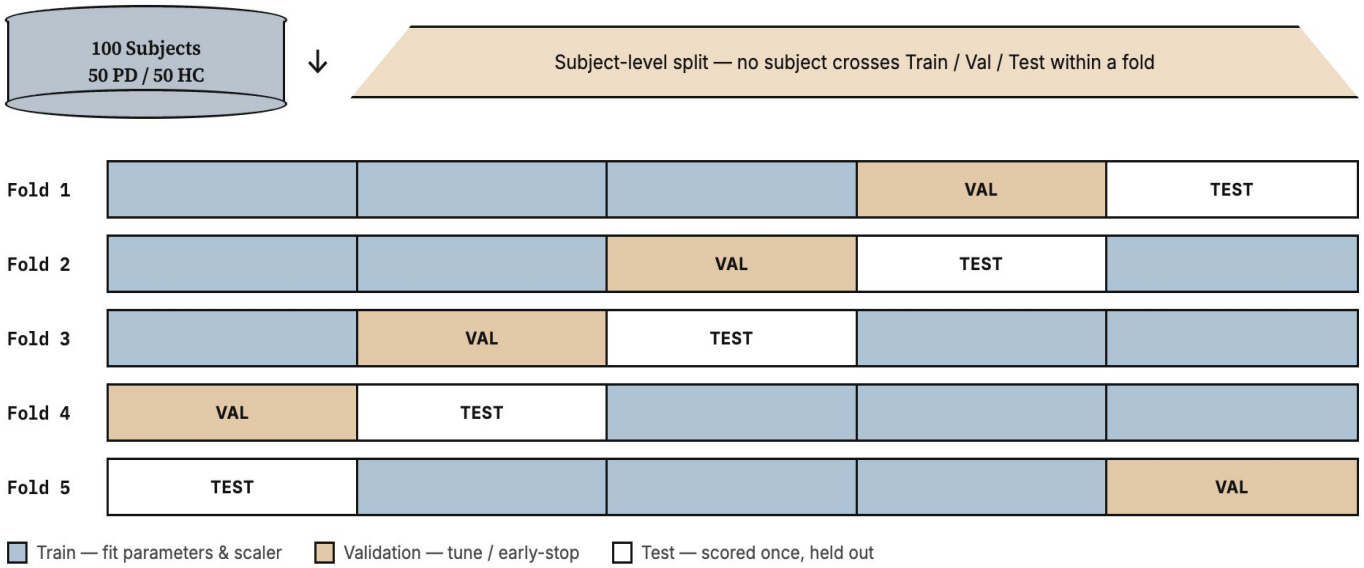
Table 1 *Mean ± SD; p from Welch's t-test (χ^2 for categorical rows).*

- PhysioNet Gait in PD
- 3 studies pooled
- about 2 min of level walking

[3] Goldberger et al., Circulation 2000

Every model is tested on people it has never seen

Materials & methods – evaluation protocol



- 60 train
- 20 validation
- 20 test subjects per fold
- test predicted once

Fig.2 *Five predefined folds: 60 train / 20 validation / 20 test subjects, class-balanced.*

Each second becomes 198 numbers

Materials & methods – proposed approach



Level	Dispersion & shape	Dynamics & spectrum
mean	standard deviation	zero-crossing rate
median	RMS	spectral energy
minimum	skewness	spectral entropy
maximum	kurtosis	

Table 2 *11 statistics per channel and second → 11 × 18 = 198 Features.*

- Mamba encoder: d_model = 128, 4 bidirectional blocks, mean + max pooling

Raw signals made Mamba collapse

Experiments & results

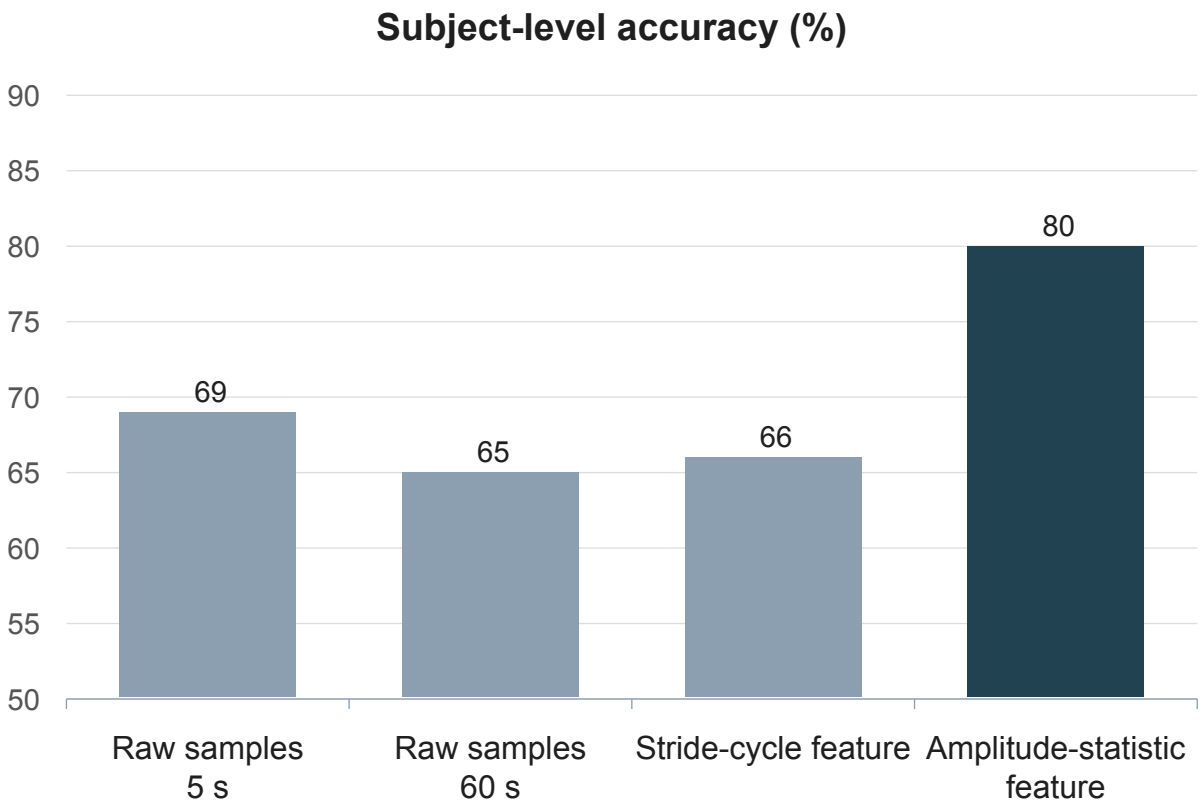


Fig. 3 *Mamba, both feet, best configuration per input (best case; reported model: 75 %).*

- Longer windows, new seeds, larger encoder: no repair
- Raw input: validation AUC 0.53 and 0.54 on folds 1 and 4

Mamba beats the deep baselines and ties the Random Forest

Experiments & results

Model (both feet)	Acc (%)	Pre (%)	F1	Sen (%)	AUC
Random Forest	73 ± 4	74 ± 4	0.73 ± 0.04	74 ± 6	0.81 ± 0.04
SVM – RBF	68 ± 5	69 ± 5	0.68 ± 0.05	66 ± 7	0.77 ± 0.05
1D-ResNet*	66 ± 5	67 ± 5	0.66 ± 0.05	56 ± 7	0.76 ± 0.05
LSTM* (same input)	63 ± 5	64 ± 5	0.63 ± 0.05	54 ± 7	0.67 ± 0.05
Mamba (ours)	**75 ± 4**	**77 ± 4**	**0.75 ± 0.04**	**86 ± 5**	**0.85 ± 0.04**

Table 3 *Subject level, 5 pooled test folds (100 subjects); mean ± SD over 1000 bootstrap resamples. *additional baselines.*

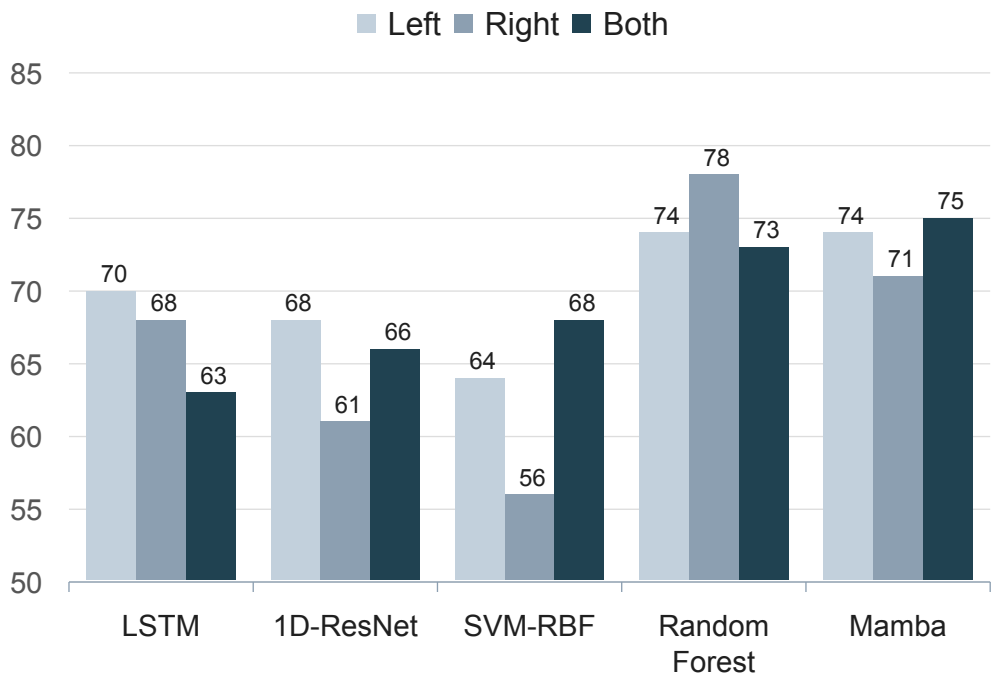


Fig. 4 *Subject-level accuracy (%) per foot configuration.*

- **Right** foot: Random Forest **78** % vs. Mamba 71 %

Why the Random Forest keeps up

Experiments & results

Reason	Evidence from the experiments
1. Small data favors ready-made features	60 training subjects per fold · best epoch 1–4 of 30
2. Same descriptors, different classifier	Mamba trained stably only on per-second statistics · causal = bidirectional (77 %, AUC 0.852)
3. The gaps are smaller than the noise	1 subject = 1 point · Random Forest: 78 % right foot → 73 % both feet · vs. Random Forest: McNemar n.s., DeLong p = 0.15
4. The ceiling is the cohort	Every model 63–75 % on both feet · no clusters in PCA / t-SNE

Table 4 *Reasons why the Random Forest keeps up.*

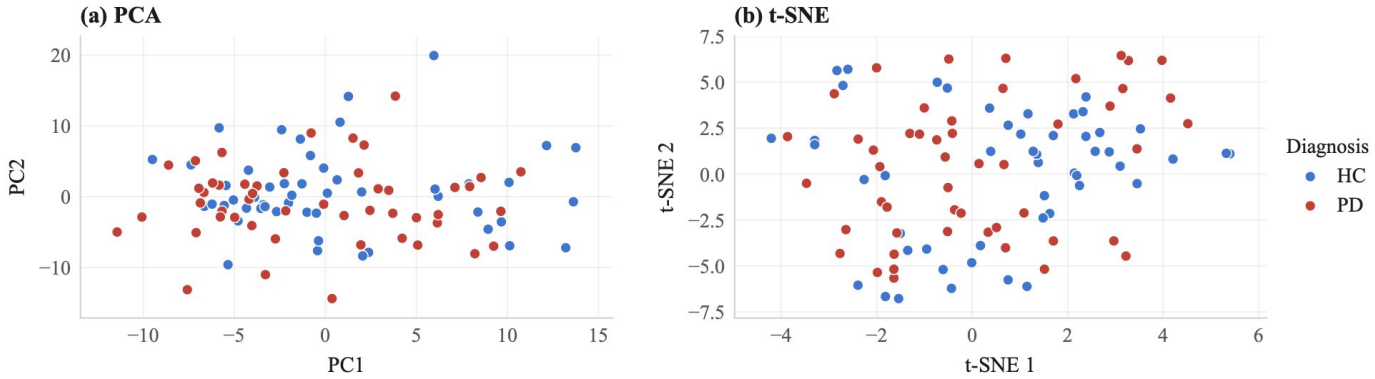


Fig. 5 *PCA/t-SNE (Unsupervised embedding of the 100 subjects)*

- Tied on 100 subjects: the data limits the comparison, not the architecture

Where does Mamba win?

Experiments & results

Mamba vs.	Δ Accuracy	Δ AUC
LSTM (same input)	**+12 (McNemar p = 0.036)**	**+0.181 (DeLong p < 0.001)**
Random Forest	+2 (n.s.)	+0.039 (DeLong p = 0.15)

Table 5 *Both feet · McNemar test for accuracy, DeLong test for AUC · same 100 subjects.*

- **Sensitivity 86** % (**Mamba**) vs. 74 % (Random Forest)

Mamba confusion matrix (rows: true, columns: predicted)

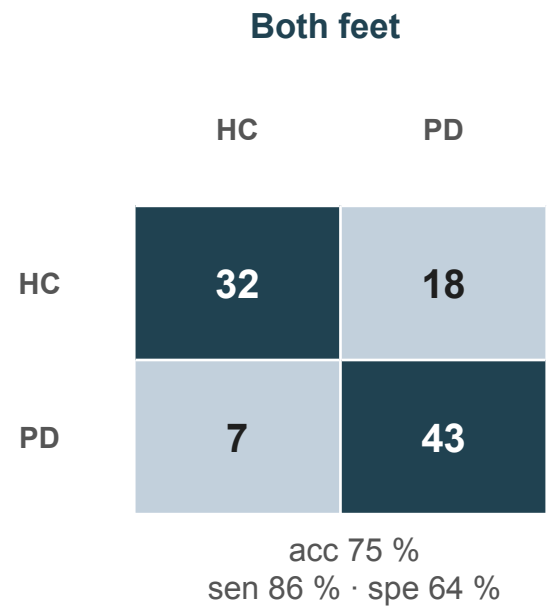


Fig. 6 *Mamba, both feet: 7 of 50 patients missed, 18 of 50 controls flagged; pooled over the five test folds.*

A window-wise split turns 73 % into 98 %

Experiments & results

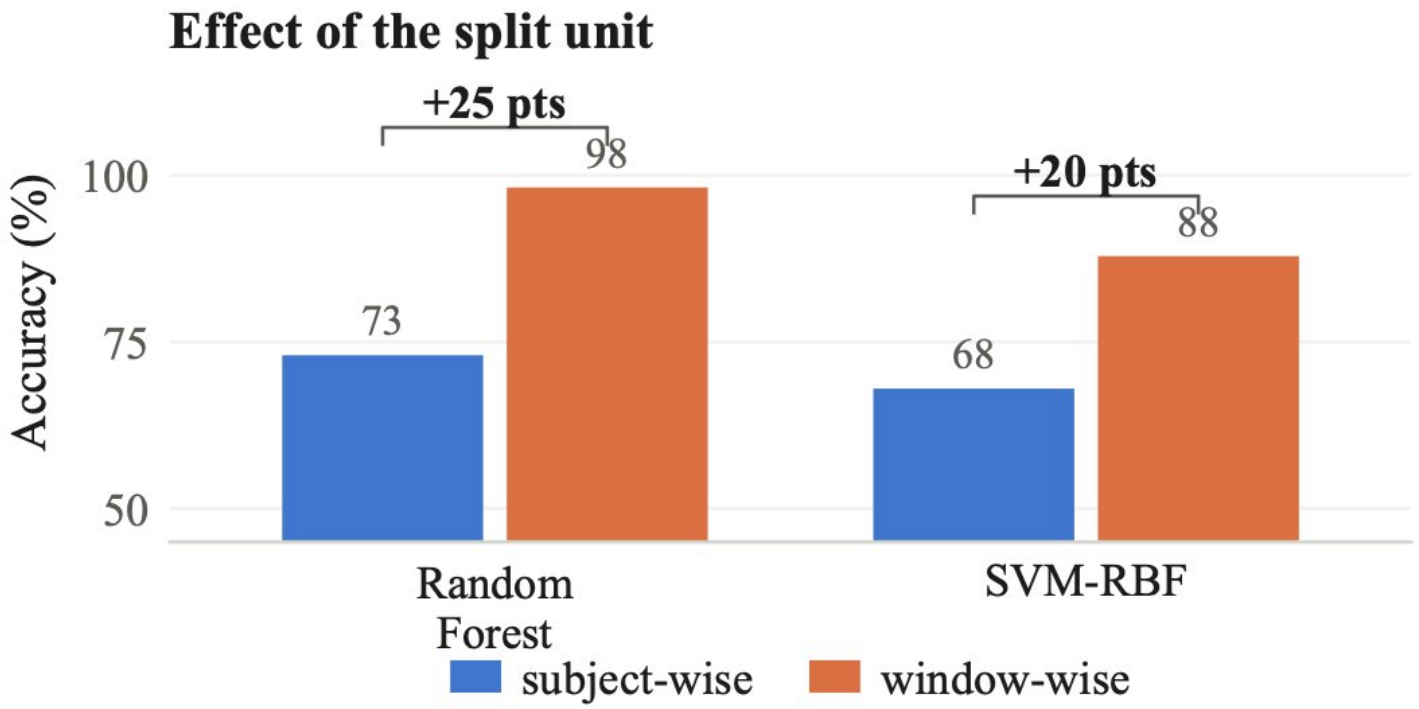


***Fig. 7** Effect of the split unit, both feet*

- Same data, features, models and code; only the split unit changed

Limitations & future work

What limits these results, and what comes next

Limitations

- Small cohort: 1 subject = 1 point (± 4 %)
- Use of temporal order not verified
- No feature-level explanation from Mamba
- Folds not stratified by study (PD rate 0.58 / 0.50 / 0.43)

Future work

- Pool the dual-task recordings and other public VGRF cohorts
- Order-shuffle test: shuffle the seconds inside each window
- Inspect which seconds Mamba's selection keeps
- Stratify the folds by study

So, can Mamba improve PD detection?

Conclusions

Answer: Partly. Mamba matches the best classical model and beats the deep baselines, but on 100 subjects it cannot significantly beat a tuned Random Forest.

1

Mamba vs. baselines

75 ± 4 % accuracy, 0.85 AUC, 86 % sensitivity; level with the Random Forest

2

What Mamba needs

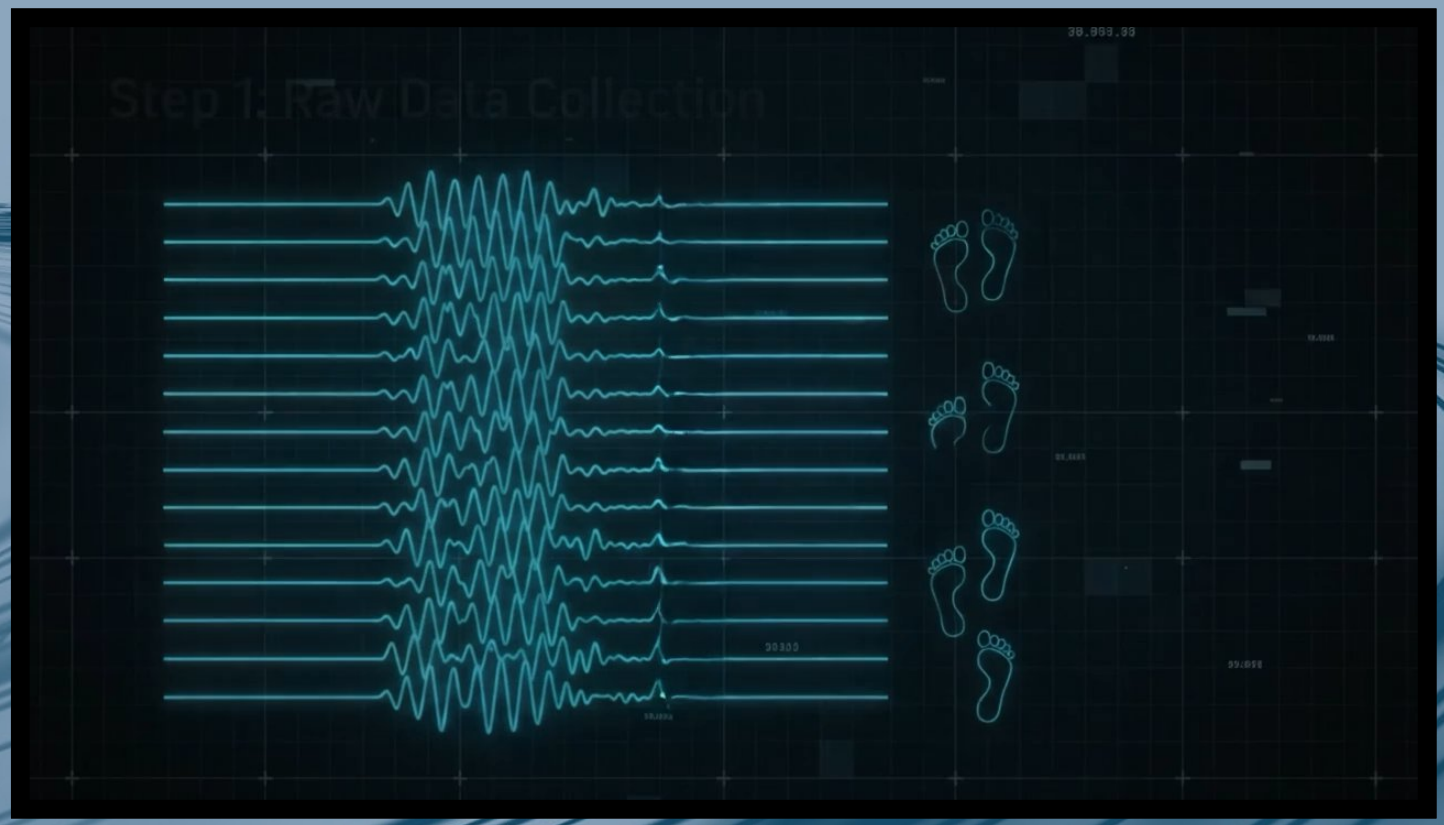
per-second statistics; raw input collapses training

3

How to evaluate

subject-wise only; a window-wise split lifts the Random Forest from 73 % to 98 %

Thank you.
Questions?



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